

Assignment 1 (ELEC 341 L1_Introduction)

Problem 1:

A temperature control system operates by sensing the difference between the thermostat setting and the actual temperature and then opening a fuel valve an amount proportional to this difference. Draw a functional closed-loop block diagram identifying the input and output transducers, the controller, and the plant. Further, identify the input and output signals of all subsystems.

error: desired temp - actual temp.

plant: room/heated element.

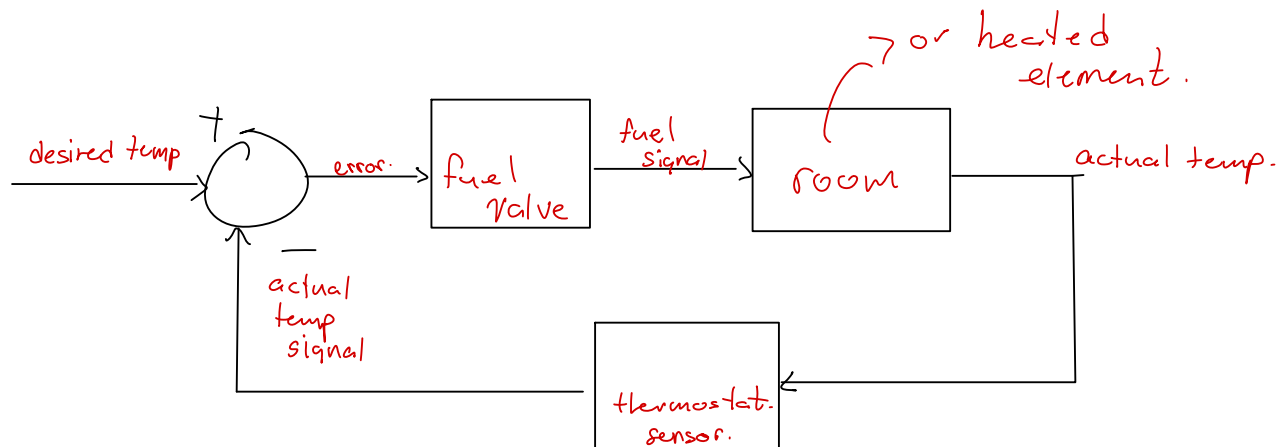
controller: fuel valve

input: desired temp.

actual output: actual temp.

output transducer: thermostat setting

input transducer: thermostat sensor.



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Problem 2:

During a medical operation an anesthesiologist controls the depth of unconsciousness by controlling the concentration of isoflurane in a vaporized mixture with oxygen and nitrous oxide. The depth of anesthesia is measured by the patient's blood pressure. The anesthesiologist also regulates ventilation, fluid balance, and the administration of other drugs. In order to free the anesthesiologist to devote more time to the latter tasks, and in the interest of the patient's safety, we wish to automate the depth of anesthesia by automating the control of isoflurane concentration. Draw a functional block diagram of the system showing pertinent signals and subsystems.

two loops.
one feedback loop
with isoflurane.

1st Output: isoflurane concentration.

2nd output: depth of unconsciousness.

plant: patient

Sensor: patient's blood pressure.

controller: anesthesiologist?

